RESPONSIBLE DIGITAL FINANCE FOR VIKSIT BHARAT:

An IKS-Informed Framework for Transforming Black-Box Credit Scoring into Responsible Borrower Explanations

Integrating Pramana, Nyaya and Dharma with Explainable AI in Digital Lending

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Abstract

Digital lending is transforming credit access, yet the decision-making process often remains opaque to borrowers. Financial institutions progressively use advanced machine-learning models to expedite loan evaluations; however, these systems typically fail to provide explanations comprehensible to the general public. Although methods such as Explainable Artificial Intelligence (XAI) and SHAP provide technical insight into the factors influencing decisions, the resulting explanations are often limited to numerical outputs or technical terminology that is not meaningful to most borrowers.

This paper describes an approach informed by Indian Knowledge Systems (IKS), integrating the principles of Pramana (valid knowledge), Nyaya (reasoning), and Dharma (responsible conduct) to enhance the clarity and justice of digital lending. Pramana confirms each explanation is grounded in verified, reliable information. Nyaya stresses a clear link between the rationale for a credit decision and its outcome, ensuring all relevant factors are accurately represented. Dharma signals the importance of explanations that are clear, respectful, and consistent with the financial system's ethical standards and regulatory systems.

Through integrating these principles, the paper proposes a three-step process: grounding explanations in strong evidence, securing clear, accurate reasoning, and communicating responsibly. The framework advocates using technology to generate decisions and technical evidence, complemented by controlled language tools that translate this evidence into borrower-accessible explanations. Rather than asserting the sufficiency of current tools such as SHAP or large language models, this approach lays the basis for future research to ensure explanations are accurate, comprehensible, and genuinely beneficial for borrowers. The overarching objective is to embed responsible and transparent explanations within digital financial services, thereby supporting skilled management and ethical leadership in the context of a developed India (Viksit Bharat).

Keywords: Indian Knowledge Systems; Pramana; Nyaya; Dharma; Explainable Artificial Intelligence; SHAP; Responsible AI; Digital Lending; Credit Scoring; Borrower Transparency